

Quiz 6

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Given the following autonomous differential equation:

$$\frac{dx}{dt} = 2x^2 - 8x.$$

Find all critical points of the given differential equation and then determine whether it is stable or unstable for each of them.

$$f(x) = 2x^2 - 8x = 2x(x-8) = 0$$

$\Rightarrow x=0$ or $x=8 \rightarrow$ two critical points

$$\begin{array}{c} x' > 0 \quad x' < 0 \quad x' > 0 \\ \xrightarrow{\quad} \quad \xleftarrow{\quad} \quad \xrightarrow{\quad} \\ \quad 0 \quad \quad 8 \end{array} \quad \begin{array}{l} \text{thus } x=0 \text{ stable} \\ x=8 \text{ unstable} \end{array}$$

Problem 2. (5 points) Solve the following initial value problem for $P(t)$:

$$\frac{dP}{dt} = 8P - 2P^2, \quad P(0) = 5.$$

and find $\lim_{t \rightarrow +\infty} P(t)$.

$$\frac{dP}{dt} = 2P(4-P)$$

$$\frac{1}{P(4-P)} = 2$$

$$\int \frac{1}{P(4-P)} dP = \int 2 dt$$

$$\int \frac{1}{4} \left(\frac{1}{P} + \frac{1}{4-P} \right) dP = \int 2 dt$$

$$\int \frac{1}{P} + \frac{1}{4-P} dP = \int 8 dt$$

$$\ln|P| - \ln|4-P| = 8t + C$$

$$\ln \left| \frac{P}{4-P} \right| = 8t + C$$

$$\Rightarrow \left| \frac{P}{4-P} \right| = e^{8t+C}$$

$$\frac{P}{4-P} = B e^{8t} \quad (B \neq 0)$$

$$P(0) = 5 \Rightarrow B = \frac{5}{4-5} = -5$$

$$\text{thus } \frac{P}{4-P} = -5 e^{8t}$$

$$\Rightarrow P = -5(4-P) e^{8t}$$

$$P = \frac{-20 e^{8t}}{1-5 e^{8t}} \Rightarrow \boxed{P = \frac{20}{5 - e^{-8t}}}$$

$$\lim_{t \rightarrow +\infty} P(t) = \lim_{t \rightarrow +\infty} \frac{20}{5 - e^{-8t}} = \frac{20}{5} = 4$$